UNIT 2 MINI-PROJECT

AGILE PROJECT PLANNING AND DEVELOPMENT

Enterprise Student Record and Academic Management Platform

Prepared by

Avtar Jassal

Roll No.: 2125UDSM1028

Course: DevOps

Based on Unit 1 Design Thinking and Product Planning Deliverables

Academic Year: 2026

1. Study of the Business Problem and Product Vision

1.1 Project Background

The Enterprise Student Record and Academic Management Platform is a web-based academic management system designed to replace fragmented paper and spreadsheet processes with a centralized digital platform. The Unit 1 work identifies duplicated data entry, delayed report generation, inconsistent attendance records, difficulty tracking academic history, and limited institution-wide visibility as major problems.

1.2 Product Vision

To become the single, trusted digital backbone for academic administration — empowering students, faculty, and administrators with accurate, real-time academic information and eliminating manual record-keeping in higher education institutions.

1.3 Mission

To design and deliver a secure, scalable, and easy-to-use platform that simplifies student registration, attendance, examinations, and reporting, so that institutions can focus on educational outcomes rather than administrative overhead.

1.4 Unit 1 Requirements Carried into Agile

Area	Agile implication
Student registration	High-priority stories for registration, validation, unique student ID and course enrollment.
Attendance	High-priority stories for faculty attendance capture, percentage calculation and student viewing.
Examinations/results	Planned after the MVP; scheduled in later sprint/release scope.
Reporting	Basic attendance/academic reports in the MVP; advanced exports and analytics later.
Access management	Role-based login for students, faculty and administrators is an MVP capability.
Quality attributes	Security, usability, reliability, scalability, performance, availability, maintainability, compliance and auditability must be considered in every sprint.

2. Agile Product Vision

2.1 Overall Product Objective

Develop a secure, user-friendly and scalable academic management platform that centralizes student records, course enrollment, attendance and basic reporting, while providing role-based access to students, faculty and administrators.

2.2 Primary Users

- Students — register for courses and view attendance and academic information.
- Faculty Members — manage class rosters, record attendance and later enter marks.
- Academic Administrators — manage institution-wide records and reports.
- Examination Cell Staff — manage examination and result workflows in later releases.
- IT Support Personnel — maintain, deploy and support the platform.
- Institution Management — review efficiency and data-quality outcomes.

2.3 Business Benefits

- Single source of truth for academic data.
- Reduced administrative turnaround time.
- Improved accuracy and auditability.
- Role-based self-service access.
- Faster attendance and reporting workflows.
- Better institutional decision-making through consolidated information.

2.4 Measurable Success Criteria

- Students can register, enroll in courses and view attendance without visiting an office.
- Faculty can mark attendance for a class session in a few minutes or less.
- Administrators can generate a basic attendance/registration report on demand.
- Registration, attendance capture and reporting are faster than the manual process.
- Data discrepancies and record-correction requests are reduced.
- The MVP is successfully released and ready for subsequent roadmap phases.

2.5 Constraints

- Stable campus internet connectivity is assumed.
- Existing data may need migration from spreadsheets or legacy systems.
- Delivery is incremental and follows the roadmap.
- Fee payment, hostel/transport and alumni management are outside the current phase.

3. Scrum Team Structure and Responsibilities

Role	Assigned responsibility for this project	Key responsibilities
Product Owner	Student representative / project requirement owner	Define and prioritize requirements; represent stakeholders; accept completed work; maintain product value.
Scrum Master	Agile process facilitator	Facilitate Scrum ceremonies; remove impediments; support collaboration; ensure Agile practices are followed.
Development Team	Design, development, testing and documentation team	Design the application; write code; test software; prepare documentation; demonstrate completed features.

3.1 Working Agreement

- All sprint work is tracked on the Scrum Board.
- User Stories must have acceptance criteria before development starts.
- Code changes are reviewed and tested before being marked Completed.
- Blockers are raised during the Daily Scrum rather than being hidden.
- The Definition of Done applies consistently to every completed story.
- Feedback from Sprint Reviews is converted into backlog items where required.

3.2 Definition of Done

- Acceptance criteria are satisfied.
- Implementation is complete and integrated with the project.
- Functional testing has been performed.
- Security and access-control checks relevant to the feature are completed.
- Documentation is updated where necessary.
- The feature is demonstrated successfully during Sprint Review.

4. Epic Documentation

Epic ID	Epic	Purpose	Main features
EPIC-01	Student Management	Manage student registration and profiles.	Registration, profile management, search, update and deletion.
EPIC-02	Course Management	Manage courses and student enrollment.	Add course, update course, assign/enroll students and view course details.
EPIC-03	Attendance Management	Digitize attendance capture and tracking.	Record attendance, view attendance and generate attendance reports.
EPIC-04	Marks and Academic Performance	Manage academic marks and performance.	Enter/update marks, calculate grades and generate result sheets.
EPIC-05	Authentication and Security	Protect platform access and sensitive academic data.	Login, logout, password reset and role-based access control.
EPIC-06	Notifications	Provide timely academic alerts.	Attendance alerts, examination notifications and result notifications.
EPIC-07	Reporting	Provide academic and administrative visibility.	Student, attendance, performance and course reports.

The seven epics follow the Unit 2 activity structure and the functional modules established in Unit 1. The MVP concentrates on registration, course enrollment, attendance, basic reporting and role-based login; examinations, advanced analytics and integrations are delivered in later roadmap phases.

5. User Story Repository

ID	Epic	User Story	Priority	SP
US-01	EPIC-05 Authentication	As a registered user, I want to log in securely so that I can access functions allowed for my role.	High	3
US-02	EPIC-01 Student Management	As a new student, I want to submit my registration details online so that my academic record can be created digitally.	High	5
US-03	EPIC-01 Student Management	As an administrator, I want mandatory registration fields validated so that incomplete records are not stored.	High	3
US-04	EPIC-01 Student Management	As an administrator, I want a unique student ID generated after registration so that each student has one authoritative record.	High	3
US-05	EPIC-02 Course Management	As a student, I want to enroll in courses offered for my program/semester so that my academic registration is recorded.	High	5
US-06	EPIC-01 Student Management	As an authorized user, I want to search and update student records so that current information is maintained.	High	5
US-07	EPIC-03 Attendance Management	As a faculty member, I want to record attendance for a class session so that attendance records remain accurate and updated.	High	5
US-08	EPIC-03 Attendance Management	As a student, I want to view my attendance record so that I can monitor my attendance.	Medium	3
US-09	EPIC-03 Attendance Management	As the system, I want to calculate attendance percentage per student per course so that attendance status is clear.	High	5
US-10	EPIC-03 Attendance Management	As an administrator, I want attendance reports by student, course or class so that academic monitoring is easier.	Medium	5
US-11	EPIC-04	As faculty, I want to	High	5

	Marks/Performance	enter and submit marks so that evaluation data can be processed.		
US-12	EPIC-04 Marks/Performance	As the system, I want to compute grades using configurable rules so that grading is consistent.	Medium	8
US-13	EPIC-04 Marks/Performance	As the examination cell, I want to schedule examinations by course so that examination activities are organized.	High	5
US-14	EPIC-04 Marks/Performance	As an administrator, I want approved results published to students so that students can access final outcomes.	High	5
US-15	EPIC-07 Reporting	As an administrator, I want academic performance reports and transcripts so that academic information can be reviewed.	High	8
US-16	EPIC-06 Notifications	As a student, I want attendance alerts so that I can act when my attendance is below the required threshold.	Medium	5
US-17	EPIC-05 Authentication	As an administrator, I want to create, update and deactivate user accounts so that access remains controlled.	Medium	5

6. Acceptance Criteria Document

Story	Feature	Acceptance Criteria
US-01	Student Login	Login page loads; registered credentials work; invalid credentials show an error; successful login redirects to the appropriate dashboard; session management functions correctly.
US-02	Student Registration	Registration form loads; mandatory details can be entered; valid submission is accepted; invalid/missing mandatory data is rejected with a useful message.
US-04	Unique Student ID	Successful registration creates exactly one student record and assigns a unique student ID.
US-05	Course Enrollment	Student can see eligible courses for the current program/semester; selected courses are recorded; duplicate/invalid enrollment is prevented.
US-07	Record Attendance	Faculty can select a course and lecture date; can mark attendance; records save successfully; saved attendance can be viewed.
US-08	View Attendance	Student can view attendance for enrolled courses; displayed information corresponds to saved attendance records.
US-09	Attendance Percentage	System calculates percentage from recorded sessions; percentage updates after attendance changes; calculation is consistent for each student/course.
US-10	Attendance Report	Authorized administrator can select student/course/class criteria and generate a readable attendance report.
US-11	Enter Marks	Authorized faculty can select the relevant course/student set, enter marks, validate required values and submit marks.
US-12	Calculate Grades	System applies the configured grading rule consistently and produces a grade for valid marks.
US-13	Exam Scheduling	Authorized examination staff can schedule an examination for a course with the required schedule information.
US-14	Publish Results	Results are publishable only after required administrative approval and are then visible to eligible students.
US-15	Performance Reports	Authorized users can generate academic performance reports/transcripts from available academic data.
US-17	User Account Administration	Authorized administrators can create, update and deactivate accounts; inactive accounts cannot authenticate.

Definition of Done: a story is complete only when its acceptance criteria are satisfied, the feature is tested, integrated, documented as needed and successfully demonstrated.

7. Prioritized Product Backlog

Rank	Story	Backlog Item	Epic	Priority	SP
1	US-01	Secure role-based login	EPIC-05	High	3
2	US-02	Online student registration	EPIC-01	High	5
3	US-03	Registration field validation	EPIC-01	High	3
4	US-04	Unique student ID	EPIC-01	High	3
5	US-05	Course enrollment	EPIC-02	High	5
6	US-07	Record attendance	EPIC-03	High	5
7	US-09	Calculate attendance percentage	EPIC-03	High	5
8	US-08	Student attendance view	EPIC-03	Medium	3
9	US-06	Search/update student	EPIC-01	High	5
10	US-10	Attendance reports	EPIC-07	Medium	5
11	US-11	Enter and submit marks	EPIC-04	High	5
12	US-17	Manage user accounts	EPIC-05	Medium	5
13	US-13	Exam scheduling	EPIC-04	High	5
14	US-14	Publish approved results	EPIC-04	High	5
15	US-15	Performance reports/transcripts	EPIC-07	High	8
16	US-16	Attendance alerts	EPIC-06	Medium	5
17	US-12	Configurable grade calculation	EPIC-04	Medium	8

7.1 Prioritization Basis

- Business value: items that directly remove the highest-impact manual work are placed earlier.
- User importance: registration and attendance are central to the MVP.
- Technical dependencies: authentication and core student records support dependent features.
- Implementation complexity: larger stories are scheduled after the foundational capabilities.

8. Story Point Estimation Sheet

Story Points	Complexity	Meaning
1	Very Easy	Small, straightforward change with minimal uncertainty.
2	Easy	Small feature with limited dependencies.
3	Moderate	Requires several implementation/test steps.
5	Difficult	Multiple components or validation paths.
8	Very Difficult	Large feature with significant dependencies or uncertainty.
13	Highly Complex	Major cross-cutting work; should normally be decomposed before implementation.

8.1 Estimation Method

The team uses Planning Poker-style collaborative estimation. Each story is discussed for scope, dependencies, technical uncertainty and testing effort. Team members independently select a point value, discuss differences, and agree on a final estimate.

8.2 Current Backlog Estimate

Total estimated backlog size: 83 story points across 17 prioritized stories.

9. Sprint Planning Document

Sprint 1

Sprint Goal: Authentication and project foundation

- US-01 Secure role-based login (3 SP)
- US-02 Online student registration (5 SP)
- US-03 Registration field validation (3 SP)
- US-04 Unique student ID (3 SP)

Planned capacity: 14 story points.

Sprint 2

Sprint Goal: Student and course management

- US-05 Course enrollment (5 SP)
- US-06 Search/update student (5 SP)

Planned capacity: 10 story points.

Sprint 3

Sprint Goal: Attendance and basic reporting

- US-07 Record attendance (5 SP)
- US-09 Calculate attendance percentage (5 SP)
- US-08 Student attendance view (3 SP)
- US-10 Attendance reports (5 SP)

Planned capacity: 18 story points.

9.1 Sprint Deliverables

Sprint	Expected increment
Sprint 1	Working project setup, authentication, registration validation and unique student record creation.
Sprint 2	Course enrollment plus student search/update functionality.
Sprint 3	Attendance capture, percentage calculation, student attendance view and basic attendance reports.

The three-sprint plan intentionally delivers the Unit 1 MVP first. Examination, advanced reporting/analytics and external integrations remain later roadmap work.

10. Sprint Backlog

Sprint	Story	Tasks	Sprint Capacity
S1	US-01	Design login flow; implement authentication; role checks; session handling; test invalid credentials.	14
S1	US-02	Create registration form, data model and submission flow.	14
S1	US-03	Validate mandatory fields and error messages.	14
S1	US-04	Generate unique student ID and persist single student record.	14
S2	US-05	Show eligible courses and save student enrollment.	10
S2	US-06	Build student search and update workflow with authorization.	10
S3	US-07	Build faculty attendance screen and save session records.	18
S3	US-09	Implement attendance percentage calculation.	18
S3	US-08	Build student attendance view.	18
S3	US-10	Generate attendance reports by supported criteria.	18

Sprint Execution Rule

Work moves from To Do → In Progress → Testing → Completed only after the relevant quality checks and acceptance criteria are satisfied.

11. Scrum Board

Backlog	To Do	In Progress	Testing	Completed
US-11 Enter marks	US-05 Course enrollment	US-07 Attendance	US-01 Login	Repository setup
US-13 Exam scheduling	US-06 Student search/update	US-09 Attendance calculation	US-02 Registration	Project structure
US-14 Publish results	US-08 Attendance view	US-10 Attendance reports	US-03 Validation	Unit 1 planning
US-15 Performance reports	US-17 Account management		US-04 Student ID	
US-16 Alerts	US-12 Grade calculation			

This is the planned/simulated Scrum Board for the Unit 2 exercise. During actual sprint execution, cards are moved as work progresses.

12. Simulated Scrum Ceremonies

12.1 Sprint Planning

- Review and clarify prioritized User Stories.
- Confirm acceptance criteria and dependencies.
- Estimate effort using story points.
- Select stories that fit sprint capacity.
- Define and communicate the Sprint Goal.

12.2 Daily Scrum

- What did I complete yesterday?
- What will I work on today?
- What challenges or blockers am I facing?

12.3 Sprint Review

The team demonstrates the completed increment to stakeholders and records feedback. For the MVP-oriented sprints, feedback should focus on ease of registration, speed of attendance marking, accuracy of attendance information and usefulness of reports.

12.4 Sprint Retrospective

Question	Simulated outcome
What went well?	Stories were prioritized around the MVP; acceptance criteria gave the team a common completion target.
What challenges occurred?	Dependencies between authentication, student records and later modules created sequencing constraints.
What should improve?	Break large stories earlier, clarify validation rules before development and keep the Scrum Board updated daily.

13. Agile Framework Comparison

Framework	Core idea	Suitable context	Assessment for this project
Scrum	Time-boxed sprints, defined roles and ceremonies	Medium teams needing structured incremental delivery	Strong fit
Kanban	Continuous flow and WIP limits	Teams with changing priorities and continuous work	Useful for support/maintenance
Extreme Programming (XP)	Engineering practices such as frequent testing and continuous feedback	Software teams needing strong engineering discipline	Useful alongside Scrum
Lean Development	Reduce waste and maximize value	Organizations optimizing process efficiency	Useful as a guiding principle
LeSS	Scale Scrum across multiple teams with a shared product	Large products with multiple Scrum teams	Later if scale requires it
SAFe	Structured scaling across teams/programs/enterprise	Large organizations with complex coordination	Too heavyweight for current project
Scrum@Scale	Scale Scrum through coordinated Scrum teams	Organizations with multiple Scrum teams	Later if team count grows

13.1 Recommended Framework

Scrum is the most suitable primary framework for the current Student Record and Academic Management Platform because Unit 2 explicitly organizes the work into Product Backlog, Sprint Planning, Sprint Backlog, Scrum Board, Daily Scrum, Sprint Review and Sprint Retrospective. XP engineering practices can complement Scrum for testing and code quality.

14. Project Timeline and Release Roadmap

Phase	Key capabilities	Delivery
Phase 1 — Foundation (MVP)	Student registration, course enrollment, attendance capture, basic reporting, role-based login	Sprints 1–3
Phase 2 — Examinations	Exam scheduling, mark entry, grading rules, result publishing and approval workflow	Subsequent sprints
Phase 3 — Reporting & Analytics	Configurable reports, transcripts, performance dashboards, exportable data	Later release
Phase 4 — Extensions & Integration	Notifications, institution ERP integration and mobile app enhancements	Later release

14.1 Release Logic

The roadmap preserves the centralized data foundation created by the MVP. Each later phase adds capabilities without changing the core product direction.

15. Burndown Chart Data

The following simulated burndown data assumes the 42 story points planned across the three MVP sprints are completed progressively. The values are a planning artifact for the Unit 2 exercise, not measured production data.

Sprint Progress Day	Remaining Story Points
Day 0	42
Day 1	40
Day 2	37
Day 3	34
Day 4	31
Day 5	28
Day 6	24
Day 7	20
Day 8	16
Day 9	12
Day 10	8
Day 11	4
Day 12	0

Interpretation

A healthy burndown should trend toward zero by the end of the planned sprint/release window. A flat period indicates a blocker or work not yet completed; a sudden drop may indicate stories being completed in batches.

16. Sprint Review Report

Sprint	Demonstrated increment	Stakeholder feedback to capture
Sprint 1	Login, role-based access, online registration, validation and unique student ID.	Confirm login usability, registration clarity, validation messages and correctness of student records.
Sprint 2	Course enrollment and student search/update.	Confirm course eligibility behavior, enrollment clarity and record-update permissions.
Sprint 3	Attendance capture, attendance percentage, student attendance view and attendance reports.	Confirm speed of faculty attendance entry, correctness of percentage calculation and usefulness of reports.

Review Outcome

The review checks whether each increment satisfies its acceptance criteria and contributes to the MVP success criteria. Any accepted change or new requirement is returned to the Product Backlog for prioritization.

17. Sprint Retrospective Report

Area	Action
Keep doing	Use clear User Stories and acceptance criteria; prioritize MVP value; keep role-based access visible in design and testing.
Improve	Split large stories earlier; refine technical dependencies before Sprint Planning; keep reporting requirements explicit.
Start doing	Use consistent test cases for each acceptance criterion; review the Scrum Board at the Daily Scrum; record blockers immediately.
Stop doing	Do not move unfinished or untested work directly to Completed; do not expand MVP scope without Product Owner prioritization.

Retrospective Commitment

The team will apply these improvements in the next sprint and review whether they reduce rework, improve predictability and make the increment easier to demonstrate.

18. Requirements-to-Backlog Traceability

Unit 1 BR	Business Requirement	Unit 2 User Stories	Release
BR-01	Single digital record per student	US-02, US-04, US-06	MVP
BR-02	Online course registration	US-05	MVP
BR-03	Daily attendance tracking	US-07, US-08, US-09, US-10	MVP
BR-04	Examination scheduling and result publishing	US-13, US-14	Phase 2
BR-05	Academic and administrative reports	US-10, US-15	MVP + Phase 3
BR-06	Role-based access control	US-01, US-17	MVP + ongoing

Functional Requirement Traceability

FR Group	Module	Stories
FR-01–FR-04	Registration and course enrollment	US-02, US-03, US-04, US-05
FR-05–FR-08	Attendance	US-07, US-08, US-09, US-10
FR-09–FR-12	Examination and results	US-11, US-12, US-13, US-14
FR-13–FR-15	Reporting	US-10, US-15
FR-16–FR-17	User/access management	US-01, US-17

19. Agile Quality, Security and Risk Plan

19.1 Non-Functional Requirements in Agile Work

Quality area	How it is handled in sprints
Security	Role-based access, secure authentication, hashed passwords/session management and authorization tests.
Usability	Responsive and simple interfaces; validate core flows with representative users.
Reliability	Error handling, validation and tests designed to avoid data loss during normal operation.
Scalability	Modular services and data model designed to support growth across academic years.
Performance	Check login, attendance marking and report generation under normal load.
Availability	Plan maintenance outside peak academic hours and design for graceful failures.
Maintainability	Modular architecture, documented code and separated backend/frontend concerns.
Compliance	Follow institutional data-retention and privacy policies.
Auditability	Maintain audit trails for critical record modifications and result-related actions.

19.2 Key Risks

Risk	Impact	Mitigation
Scope creep beyond MVP	High	Product Owner prioritizes backlog and protects sprint goal.
Incomplete legacy data migration	Medium	Validate source data and plan migration before dependent releases.
Changing institutional policies	Medium	Keep business rules configurable where practical and refine backlog.
Authentication/access defects	High	Test role permissions and invalid access paths in every relevant sprint.
Incorrect attendance calculations	High	Use explicit acceptance criteria and test calculations with representative data.

20. Agile Alignment with System Architecture

The Unit 1 architecture uses a layered structure: Presentation Layer, API/Application Layer, Business Logic Layer, Data Access Layer, Database Layer, with external integrations planned for later phases.

Layer	Agile work alignment
Presentation Layer	Student portal, faculty portal and admin console stories are developed incrementally.
API / Application Layer	Authentication, authorization and routing support multiple feature stories.
Business Logic Layer	Registration, attendance, examinations/results and reporting can evolve as separate modules.
Data Access Layer	Validation, data access and caching support reliable feature implementation.
Database Layer	Stores student, course, attendance and result data as the centralized source of truth.
External Integrations	Notification, ERP and payment integrations remain later-roadmap work.

Architecture Diagram — Text Representation

Presentation Layer

↓

API / Application Layer

↓

Business Logic Layer

↓

Data Access Layer

↓

Database Layer

↓

External Integrations (later phases)

21. Unit 2 Completion Checklist

No.	Required Unit 2 Deliverable	Status
1	Agile Project Planning Report	Completed
2	Product Vision Document	Completed
3	Scrum Team Structure	Completed
4	Scrum Roles and Responsibilities Document	Completed
5	Epic Documentation	Completed
6	User Story Repository	Completed
7	Acceptance Criteria Document	Completed
8	Prioritized Product Backlog	Completed
9	Story Point Estimation Sheet	Completed
10	Sprint Planning Document	Completed
11	Sprint Backlog	Completed
12	Scrum Board	Completed
13	Burndown Chart	Completed
14	Sprint Review Report	Completed
15	Sprint Retrospective Report	Completed
16	Agile Framework Comparison Report	Completed
17	Project Timeline and Release Roadmap	Completed
18	Complete Agile Documentation	Completed

Final Project Statement

This Unit 2 Agile Mini-Project translates the Unit 1 Enterprise Student Record and Academic Management Platform into an actionable Agile delivery plan. The backlog, user stories, acceptance criteria, story points, sprint plan, Scrum board, review, retrospective and roadmap remain aligned with the original product vision, business requirements, functional requirements, MVP and architecture.

22. Source Alignment and Document Notes

Primary Source Documents Used

- Unit 1 Mini Project — Enterprise Student Record and Academic Management Platform: Design Thinking and Product Planning — Consolidated Deliverables.
- Unit 2 Agile Mini-Project — Student Activities and Student Deliverables.

Consistency Notes

- The project name and product direction are kept consistent with Unit 1.
- The Unit 2 epics, Scrum roles, user-story format, acceptance criteria, backlog, story points, sprints, Scrum board, ceremonies and framework comparison are applied to the same platform.
- The MVP scope remains registration, course enrollment, attendance, basic reporting and role-based login; examination/result workflow and advanced analytics are treated as later phases.
- The simulated burndown values are clearly identified as planning data rather than actual development measurements.

Prepared for academic submission.